

24 Rigorous Limit Proofs

Ultimate Calculus GPA Saver



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The 4 BIG Definitions

$$\lim_{x \rightarrow a} f(x) = L \text{ means } \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

$$\lim_{x \rightarrow \infty} f(x) = L \text{ means } \forall \varepsilon > 0, \exists N > 0 \text{ s.t. } x > N \Rightarrow |f(x) - L| < \varepsilon$$

$$\lim_{x \rightarrow a} f(x) = \infty \text{ means } \forall M > 0, \exists \delta > 0 \text{ s.t. } 0 < |x - a| < \delta \Rightarrow f(x) > M$$

$$\lim_{x \rightarrow \infty} f(x) = \infty \text{ means } \forall M > 0, \exists N > 0 \text{ s.t. } x > N \Rightarrow f(x) > M$$

definitions adapted from Single Variable Calculus by James Stewart

(Q1.) $\lim_{x \rightarrow 4} (2x - 3) = 5$

(Q2.) $\lim_{x \rightarrow 5} \frac{1}{x - 3} = \frac{1}{2}$

(Q3.) $\lim_{x \rightarrow \infty} \frac{1}{x - 3} = 0$

(Q4.) $\lim_{x \rightarrow 3^+} \frac{1}{x - 3} = \infty$

(Q5.) $\lim_{x \rightarrow \infty} \frac{3x}{2x + 1} = \frac{3}{2}$

(Q6.) $\lim_{x \rightarrow 2} 3x^2 = 12$

(Q7.) $\lim_{x \rightarrow \infty} \sqrt{3x + 1} = \infty$

(Q8.) $\lim_{x \rightarrow 5} \sqrt{3x + 1} = 4$

(Q9.) $\lim_{x \rightarrow -2} (-3x + 5) = 11$

(Q10.) $\lim_{x \rightarrow 3} \frac{x}{x - 2} = 3$

(Q11.) $\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$

(Q12.) $\lim_{x \rightarrow \infty} \sqrt{x - 3} = \infty$

$$(Q13.) \lim_{x \rightarrow 2} x^3 = 8$$

$$(Q14.) \lim_{x \rightarrow 6} \left(\frac{1}{2}x + 1 \right) = 4$$

$$(Q15.) \lim_{x \rightarrow \infty} \frac{x}{x^2 - 1} = 0$$

$$(Q16.) \lim_{x \rightarrow \infty} (x^2 + 2x) = \infty$$

$$(Q17.) \lim_{x \rightarrow 1} (x^2 + 2x) = 3$$

$$(Q18.) \lim_{x \rightarrow 1^+} \frac{x}{2x - 2} = \infty$$

$$(Q19.) \lim_{x \rightarrow \infty} \frac{x^2}{3x^2 + 1} = \frac{1}{3}$$

$$(Q20.) \lim_{x \rightarrow 2^+} \frac{1}{3x - 6} = \infty$$

$$(Q21.) \lim_{x \rightarrow -2} (x^2 + x + 3) = 5$$

$$(Q22.) \lim_{x \rightarrow \infty} \frac{x}{2x - 8} = \frac{1}{2}$$

$$(Q23.) \lim_{x \rightarrow 3} \frac{1}{(x - 3)^2} = \infty$$

$$(Q24.) \lim_{x \rightarrow \infty} \frac{x^2}{x + 1} = \infty$$

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